Supplementary Material: Fixed Points and Certified Boundedness of Unrolled Born/Contrast-Source Inverse Scattering Networks

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1 Contrast-solve truncation sweep

The contrast step uses a fixed $n_{\mathrm{cg}} = 15$-step conjugate-gradient loop. The certified blocks C and D are closed-form and touch no iterative solve; the contrast-path blocks A, B and the full-layer norm that contains them do.

With the implemented n_{cg} the relative residual of the regularised system is 2.90×10^{-4}; it falls to 8.18×10^{-10} at $n_{\mathrm{cg}} = 200$. The field block D contains no iterative solve, and there the analytic directional derivative and a central difference agree to 8.6×10^{-13}. On the contrast-path block A at the implemented truncation the central difference and the analytic directional derivative disagree by 0.82; the relative discrepancy does not shrink as the finite-difference step is refined, which is the signature of a difference quotient that does not converge to the derivative at that truncation. Converging the solve restores the agreement to 1.79×10^{-2}.

Table 1: Contrast-solve truncation sweep at a fixed operating point, $n_{\mathrm{cg}} \in \{5, \ldots, 200\}$, float64. C, D are bit-identical over the whole sweep; the contrast-path blocks A, B and the full-layer norm that contains them are not, and their min and max are tabulated.

quantity	min	max	behaviour over the sweep
$\sigma_{\max}(C)$ (CG-free)	0.8480	0.8480	bit-identical
$\sigma_{\max}(D)$ (CG-free)	0.7935	0.7935	bit-identical
$\sigma_{\max}(A)$ (contrast)	0.723	0.874	varies
$\sigma_{\max}(B)$ (contrast)	0.118	3.515	varies
$\sigma_{\max}(DT_k)$ (full layer)	0.7368	3.6769	varies, non-monotonically

The full-layer norm mixes the two CG-free blocks with the two solver-dependent ones, so it inherits the artefact even though its own closure has no solver in it. It is reported because a reader who wants one number per layer will reach for it, and it is never certified: a statement about $\sigma_{\max}(DT_k)$ would be a statement about the truncation the trained weights happen to be calibrated to.

2 First-layer gradient norm

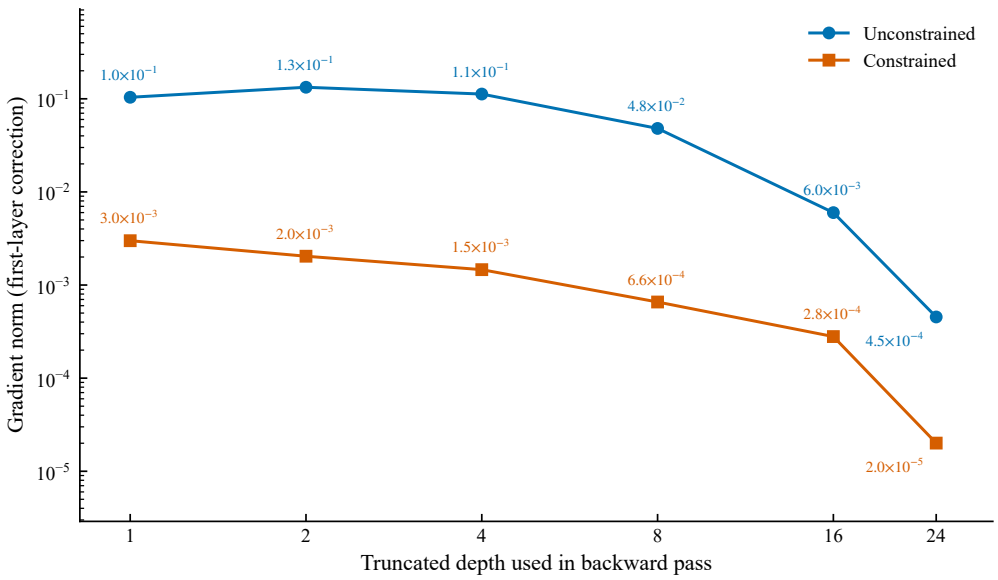


Figure 1: First-layer gradient norm versus the truncated depth used in the backward pass.